

Lab 14.1 - Back Up, Destroy, Restore

KVM Virtualisation on Oracle Linux - Day 2, Module 14 (Backup & Disaster Recovery)

Overview

Module 6 was explicit that a snapshot is not a backup - it lives inside the same qcow2 file as the VM it protects, so anything that destroys the VM's storage destroys the snapshot right along with it. This lab builds the real thing: an external backup, copied outside libvirt's own storage entirely, that survives even the most destructive command in this course - `virsh undefine --remove-all-storage`. You'll prove it by deliberately causing that exact disaster, then rebuilding `ol-lab-01` purely from the backup files, with its data verified intact afterwards.

Note: Backups in this lab are stored on the `/mnt/data` volume from Lab 5.1, not the boot volume - a real backup should never live on the same disk as the thing it's protecting, and this also keeps the boot volume's limited free space clear.

Learning Objective

By the end of this lab, you will be able to:

- Explain why an external backup survives events a snapshot cannot
- Back up a VM's XML definition, disk, and any attached seed media to a separate location
- Simulate a genuine disaster by completely destroying a VM's definition and storage
- Restore a VM purely from backup files, with no dependency on any state libvirt itself retained
- Verify data integrity after restore, not just that the VM happens to boot

Step by Step Guide

Prerequisite: Have `ol-lab-01` Running with Some Recognisable Data Inside

1. Connect to your lab host, then find and connect to `ol-lab-01`:

```
ssh opc@<your-lab-instance-ip>
```

```
sudo virsh domifaddr ol-lab-01
```

```
ssh -i ~/.ssh/id_ed25519 cloud-user@<ol-lab-01's address>
```

2. Write a recognisable, timestamped marker to the /data disk from Lab 5.1 - this is what Step 4 will check for after the disaster:

```
echo "Backup drill $(date)" | sudo tee /data/backup-marker.txt  
cat /data/backup-marker.txt
```

Expected result: The marker file exists with today's date and time - note the exact content, since Step 4 confirms it survived byte for byte.

Step 1: Shut It Down; Save Its XML and Disk to a Backup Location

Back on the host, capture everything needed to fully reconstruct ol-lab-01 later - not just its disk, but its definition and any attached seed media too.

3. Shut the VM down, confirming it has genuinely stopped:

```
sudo virsh shutdown ol-lab-01  
watch -n 2 sudo virsh list --all  
# Ctrl-C once state shows "shut off"
```

4. Create a backup location on the dedicated data volume, and save the XML definition:

```
sudo mkdir -p /mnt/data/backups/ol-lab-01  
sudo virsh dumpxml ol-lab-01 > /tmp/ol-lab-01.xml  
sudo cp /tmp/ol-lab-01.xml /mnt/data/backups/ol-lab-01/ol-lab-01.xml
```

5. Copy the disk and its seed ISO to the same location:

```
sudo cp /var/lib/libvirt/images/ol-lab-01.qcow2 \  
/mnt/data/backups/ol-lab-01/ol-lab-01.qcow2  
sudo cp /var/lib/libvirt/images/ol-lab-01-seed.iso \  
/mnt/data/backups/ol-lab-01/ol-lab-01-seed.iso
```

6. Confirm everything backed up correctly:

```
sudo ls -lh /mnt/data/backups/ol-lab-01/
```

Expected result: Three files listed - the XML definition, the qcow2 disk, and the seed ISO - each with a non-zero size matching the originals.

Note: The XML definition matters as much as the disk here - it's what tells libvirt about ol-lab-01's vCPU count, memory, network attachment, and the exact path to its seed ISO. A disk backup alone is not a complete backup.

Step 2: Simulate Disaster - undefine --remove-all-storage

Warning: This step is genuinely destructive and cannot be undone through libvirt itself - that's the entire point of the drill. Do not run this against any VM you have not already backed up in Step 1.

7. Destroy ol-lab-01 completely - definition and every disk it references:

```
sudo virsh undefine ol-lab-01 --remove-all-storage
```

Expected result: Domain ol-lab-01 has been undefined, with associated storage volumes listed as removed.

8. Confirm the disaster is real, not just simulated:

```
sudo virsh list --all
```

```
sudo ls /var/lib/libvirt/images/ol-lab-01.qcow2
```

Expected result: ol-lab-01 no longer appears in virsh list --all at all, and the ls command reports No such file or directory - the VM and its original disk are genuinely gone from libvirt's own storage.

Step 3: Restore - Disk Back, Define the XML, Start

Rebuild ol-lab-01 purely from the backup files created in Step 1 - nothing here depends on libvirt having retained any memory of the VM.

9. Copy the disk and seed ISO back into place:

```
sudo cp /mnt/data/backups/ol-lab-01/ol-lab-01.qcow2 \  
/var/lib/libvirt/images/ol-lab-01.qcow2
```

```
sudo cp /mnt/data/backups/ol-lab-01/ol-lab-01-seed.iso \  
/var/lib/libvirt/images/ol-lab-01-seed.iso
```

10. Redefine the VM from the backed-up XML:

```
sudo virsh define /mnt/data/backups/ol-lab-01/ol-lab-01.xml
```

Expected result: Domain ol-lab-01 defined from /mnt/data/backups/ol-lab-01/ol-lab-01.xml.

11. Start it:

```
sudo virsh start ol-lab-01
```

```
sudo virsh list --all
```

Expected result: ol-lab-01 listed again, state running - fully reconstructed from files that exist entirely independently of anything libvirt itself remembered.

Step 4: Confirm the Guest Boots with Its Data Intact

12. Find the restored VM's address and connect:

```
sudo virsh domifaddr ol-lab-01
```

```
ssh -i ~/.ssh/id_ed25519 cloud-user@<ol-lab-01's address>
```

13. Confirm the marker file from the Prerequisite step survived the disaster, byte for byte:

```
cat /data/backup-marker.txt
```

Expected result: The exact same timestamped content written before Step 1 - proof the restore recovered genuine data, not just a VM that happens to boot.

Note: Notice this VM did not go through cloud-init's first-boot sequence again, unlike every fresh deployment in Labs 3.1 and 13.1 - the restored disk still carries cloud-init's own record of having already run for this exact instance-id, so it correctly recognised itself as the same machine rather than a new one. This is a meaningfully different outcome from Lab 13.1's templating workflow, where a fresh identity was exactly what was wanted - here, preserving the original identity is the correct behaviour.

Outcomes

By the end of this lab, you have a genuine, working backup and restore cycle: ol-lab-01 was completely destroyed - definition and storage both - and rebuilt entirely from files that lived outside libvirt's own storage the whole time, with its data confirmed intact afterwards. This is the capability Module 6 explicitly said snapshots do not provide, and the difference matters - a snapshot would have been destroyed in the same `undefine --remove-all-storage` command that this lab's backup survived without issue.